Individual Project

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Due data: 21/10/2022

Introduction:

Background: Nowadays, the weather is changing more and more frequently, people's travel is becoming more and more convenient as well, so I developed a website that provide to users can check every city's weather, today's weather details and some forecast for next seven days. The information will be returned as some numeric data and weather images to users.

Because of the importance of weather to human beings, it is very important that people are always aware of the weather around them. When people have a specific trip, they inevitably need to know the weather details of their destination, even it is more easier to allowe them to better plan their luggage that they should carry.

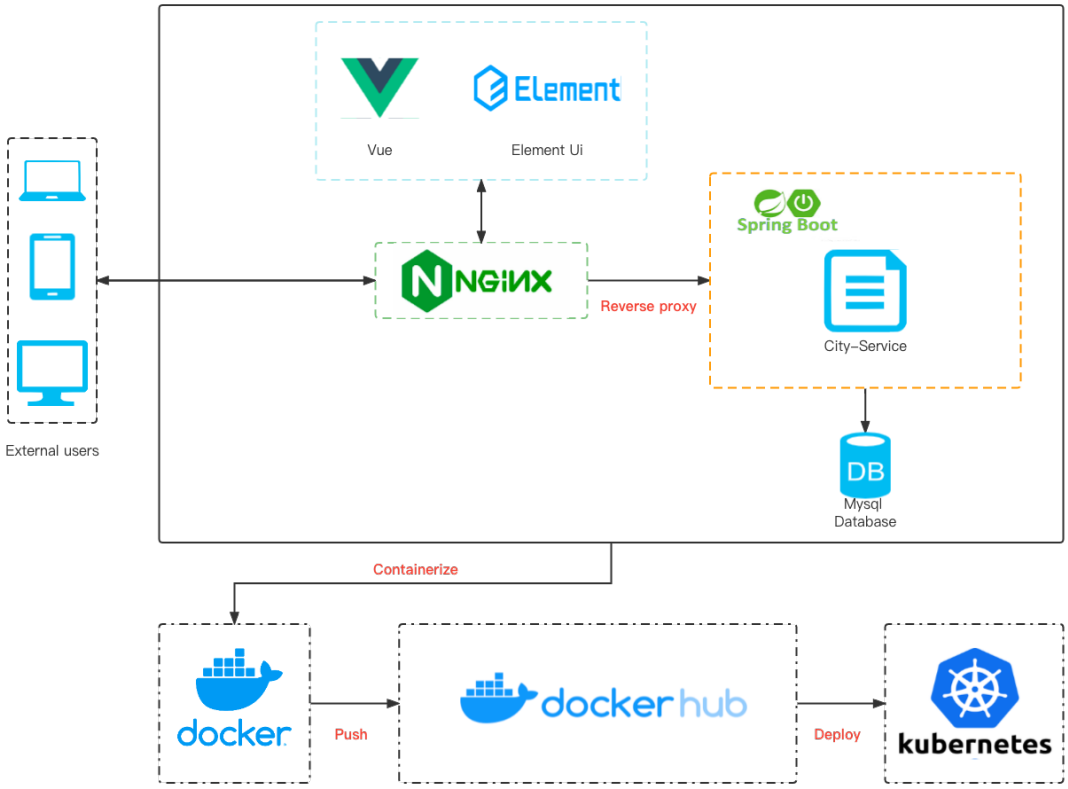
This project's objective is to return the weather in specific city which include current temperature, humidly and forecast. Which will give user enough data to help them decide in different weather. This website will establish on a Kubernetes cluster by google cloud platform, it can accept large data volume access requests, no program downtime due to the high concurrency. Generate different dynamic weather pictures according to different weather.

For traditional computing solution, some of those website or app will store the data locally, they may need to spend lots of cost on their database. If the server goes down for different reasons, all websites will be affected and the scalability for traditional computing solution is worse than cloud computing as well. Because there are so many factors that can affect the weather, there will be a large amount of

different kinds of data being observed, resulting in the need for frequent expansion of the database, that is the limitation of traditional computing.

However, for cloud computing, which has more benefits (eg. reduced investments and proportional costs, increased scalability, increased availability and reliability). For frequent data storage and updates, database size can be easily expanded by purchasing from service providers. My database will not store observations to speculate on the weather, rather than displays the results to the user from existing API results. In extreme cases, the serve will not shutdown suddenly because the availability and reliability.

Architecture Design:



Technical Solutions:

My front-end is using Vue to build, and Element-UI to render the page and use Nginx to proxy front-end and back-end. Users can visit my website from port

80, from the reverse proxy that provided by Nginx, the data from back-end can be passed to front-end in port 8080 and my back-end is contributed by using Springboot framework, MySQL as my database to store cities' name and their longitude and latitude. After building up my front-end and back-end, I use docker to containerize my application with multiple containers running in a microservice architecture, and push them into my docker hub repository to store them which is also convenient to orchestrate into Kubernetes. Kubernetes will distribute all applications into different pods which is also the smallest unit in Kubernetes, for the existing pod Kubernetes will generate to a deployment and service can let outside world visit those applications, you can decide how many pods or replicas will be generate by yourself.

If we choose to use three nodes to create a Kubernetes cluster (2 vCPU + 4 GB memory) and 100GB disk on google cloud platform, monthly cost would be US$158.38.

Estimated monthly cost PREVIEW

US$158.38

That's about US$0.22 per hour

Pricing is based on the resources you use, management fees, discounts and credits. Learn more

US-CENTRAL1

Item	Estimated monthly cost
Cluster management fee	US$73.00
US$0.10/hour x 730 hours	
default-pool (e2-medium)	
2 vCPU + 4 GB memory (US$24.46 x 3 nodes)	US$73.38
100 GB standard persistent disk (US$4.00 x 3 nodes)	US$12.00
Cluster discounts	
Monthly free tier credit	Not included
Estimated monthly cost	**US$158.38**